PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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1. Project Overview

- **Problem Statement:** The modern Food & Beverage (F&B) industry operates in a constant state of high-entropy chaos, characterized by razor-thin profit margins and daily financial leakage. This leakage stems from two primary vectors: Back-of-House (BoH) inventory spoilage and Front-of-House (FoH) unoptimized table turnover. Traditional Point-of-Sale (POS) and inventory systems fail because they are merely static digital ledgers; they require human intervention to act as the "bridge" between an external shock (e.g., a supplier texting a sudden price hike via WhatsApp) and internal adaptation (e.g., updating a live menu to protect margins).

- **Target Domain:** Independent Restaurant Owners , Restaurant Operations Managers , and Procurement Managers in F&B

- **Proposed Solution Summary:** AutoYield acts as an AI-native operations kernel powered by Z.AI's GLM. By shifting the paradigm from manual data entry to autonomous operational execution, AutoYield ingests unstructured signals—ranging from blurry supplier invoices to messy handwritten dockets. It utilizes an advanced "Dual-Agent Debate" architecture to dynamically resolve the eternal conflict between maximizing immediate profit and maintaining long-term customer satisfaction. The system ultimately executes real-time, zero-latency menu and inventory adjustments across the entire restaurant ecosystem, effectively replacing manual micromanagement with an autonomous digital brain.

2. Background & Business Objective

- **Background of the Problem:** For decades, restaurant management software has been inherently reactive. Currently, a massive operational disconnect exists between the kitchen and the floor. The kitchen naturally desires to clear expiring inventory to reduce overhead waste (Cost Reduction), while the floor staff is motivated to serve the highest quality, most popular items to secure positive reviews and tips (Customer Experience). When an anomaly occurs—such as a sudden 20% spike in the cost of salmon communicated casually via text—the information is typically lost in an "unstructured data blindspot". The static system cannot process this text, the menu remains unchanged, and the restaurant silently absorbs the financial loss until end-of-month accounting.

- **Importance of Solving This Issue:** Addressing this gap accelerates operational response times from days (or weeks) to literal seconds. By bridging the gap between the storage fridge and the customer's digital ordering interface, restaurants can salvage thousands of Ringgit previously lost to food spoilage and missed margin opportunities. It eliminates the cognitive load on restaurant managers, allowing them to focus on hospitality rather than spreadsheet administration.

- **Strategic Fit / Impact:** AutoYield aligns perfectly with the future of AI-Native enterprise software by functioning as an "autopilot" rather than a mere "copilot." While competitors offer predictive dashboards that require humans to click "execute," AutoYield orchestrates the fix directly through the operational kernel. This creates a highly defensible technological moat that transforms raw operational chaos into streamlined, autonomous profitability.

3. Product Purpose

3.1. Main Goal of the System

To provide a proactive, autonomous AI "brain" capable of maximizing F&B yield through real-time, constraint-aware operational adjustments. The system seeks to act as an automated General Manager, digesting market variables and internal constraints to output mathematically optimized business decisions without human bottlenecks.

3.2. Intended Users (Target Audience)

- **Restaurant Owners & Founders:** Seeking to maximize gross margins, prevent financial leakage, and gain high-level oversight over autonomous system decisions.
- **F&B Operations Managers:** Needing to reduce the daily cognitive load of balancing supplier negotiations, stock counts, and sudden menu changes.
- **Head Chefs & Kitchen Managers:** Requiring dynamic synchronization between their actual fridge stock and the active orders hitting their Kitchen Display System (KDS).

4. System Functionalities

4.1. Description

AutoYield operates as a continuous, closed-loop economic engine. The system models the restaurant's daily operation as a dynamic optimization problem: Profit = f(inventory_levels, supplier_costs, live_demand, customer_order). By integrating unstructured data processing with relational state management (FastAPI + Supabase), the system perceives market shocks, reasons through potential solutions, checks physical kitchen constraints, and directly executes database mutations to update the restaurant's reality in real-time.

4.2. Key Functionalities

- **Unstructured Asset Ingestion (The "Interpreter"):** Translates messy, human-language inputs (e.g., OCR-extracted invoices, WhatsApp texts) into structured operational data.

- **Dual-Agent Decision Kernel:** Triggers a debate between the P-Agent (Profit/Margin Maximizer) and the R-Agent (Risk/Reputation Protector) in three scenarios:
    - User chat messages containing price changes, promotions, or discounts
    - Invoice price spikes above 20%
    - Backend-monitored inventory crises or order surges. The debate resolves the optimal tactical response.

- **Proactive Auto-Triggering (The "Healer"):** A real-time control console designed for demonstrations. It allows operators to manually inject extreme business shocks: adjust order generation speed, multiply inventory quantity or unit cost (globally or per item), and simulate oil price spikes. The interface visually exposes the AI's internal debate trace as it scrambles to self-heal the business.

- **"God-Mode" Sandbox:** A specialized, real-time control console designed for hackathon judges to manually trigger extreme crises (e.g., slashing inventory by 50% or spiking oil prices). This interface visually exposes the AI's internal thought trace as it scrambles to self-heal the business.

- **Human Authorization & Clarification Panel:** An adaptive safety net. The system pauses autonomous execution for high-stakes actions (e.g., unusually large spend or excessive price changes) or when invoice data is missing critical fields. It generates structured requests for human approval or missing information, ensuring that important decisions remain supervised.
- **Proactive Crisis Loop:** A background monitor checks inventory and the active order queue every few seconds. When stock drops below its minimum level or kitchen load exceeds a safe threshold, it automatically activates the crisis response workflow without any user input.

4.3. AI Model & Prompt Design

4.3.1. Model Selection

The core cognitive engine utilizes the Z.AI GLM. This model was specifically selected due to its superior capability in handling complex, multi-constraint reasoning tasks. Furthermore, Z.AI GLM excels at strict role-prompting (crucial for our Dual-Agent debate) and consistently outputs reliable JSON structures necessary for executing database updates via our FastAPI backend.

4.3.2. Prompting Strategy

The system employs a Multi-Agentic Role-Playing Strategy. Hard-coded logic trees fail in restaurant environments due to extreme variability. Instead, we prompt the P-Agent with a strict persona focused solely on defending margins (e.g., "Raise prices immediately to recover costs"). Conversely, the R-Agent is prompted with historical customer retention data (e.g., "High prices cause churn; let's bundle items instead"). A final "Resolution Prompt" synthesizes these arguments against current kitchen capacity constraints to select the optimal path. This strategy perfectly mirrors real-world executive decision-making.

4.3.3. Context & Input Handling

To manage input constraints and prevent context window overflow from massive, messy OCR data dumps, the system utilizes a targeted extraction mechanism. The Unstructured Ingestor is prompted to extract only the operational intent, key entities (item, price, date), and sentiment. If a

user uploads an oversized multi-page invoice, the system truncates non-essential descriptive text, chunking only the numerical line items to ensure rapid, cost-effective inference.

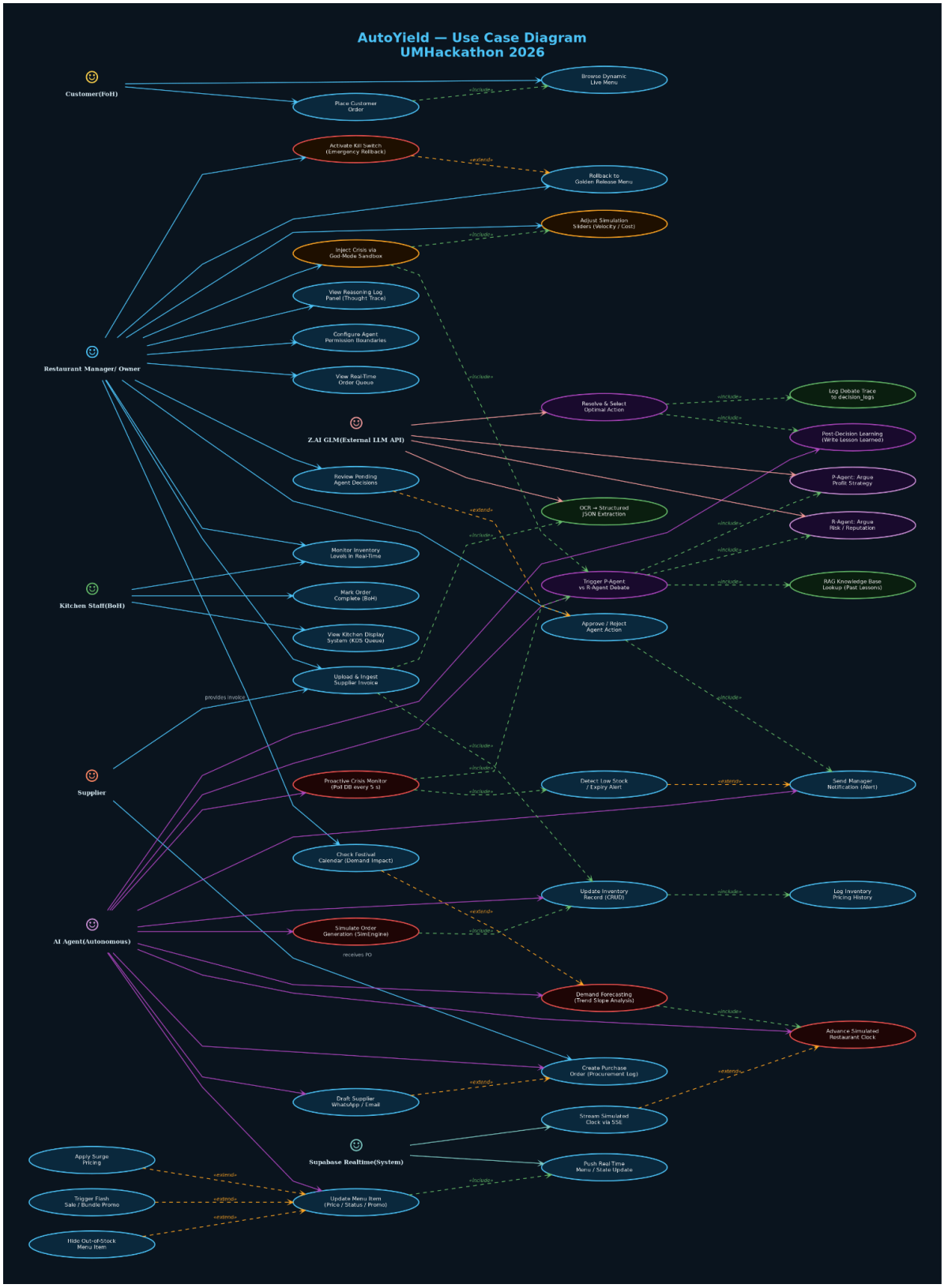
4.3.4. Fallback & Failure Behavior

If the Z.AI GLM generates a tool call that fails Pydantic schema validation (e.g., missing required fields) or the backend permission service detects a violation (e.g., purchase order > RM500 or price change > 15%), the system returns a failed or pending_approval status. For permission-based violations, a structured notification is sent to the Human Authorization Panel. For schema-level errors, the action is rejected with an explicit error message; no automatic retry is attempted.

5. User Stories & Use Cases

- **User Stories:**
  - As an Operations Manager, I want to upload a blurry image of a handwritten supplier invoice so that the system can automatically extract the data, detect a 35% price spike in salmon, and immediately suggest compensatory menu adjustments.
  - As a Kitchen Chef, I want the system to automatically rewrite the digital frontend menu when my rice supply is depleted, redirecting customer demand to noodle-based dishes without me having to leave my station.

- **Use Cases (Main Interactions):**
  - **Inventory Monitoring Flow:** The system detects stock nearing its expiry date. It triggers the Dual-Agent Debate, executes a promotional strategy, and updates the Live Menu. It then assesses remaining stock and autonomously drafts a resupply order via WhatsApp to the primary vendor.
  - **Ambiguity / Clarification Flow:** An unstructured input (invoice) is scanned. The AI detects missing unit prices. The process halts, moving to the Human Decision Panel for administrative input, ensuring data integrity.
  - **Demand Forecasting & Customer Ordering Strategy:** The GLM analyzes order history combined with external contexts (time, day, weather) to predict an evening demand surge. It proactively tightens reorder triggers and adjusts dynamic pricing to maximize yield before a shortage occurs.
- **Use Case Diagram:**



Use Case Diagram

6. Features Included (Scope Definition)

- **Conversational Chatbot Input Area:** Supports natural language text and image uploads (invoices/dockets) for unstructured ingestion.

- **Real-time Order Queue Visualizer & Business Dashboard:** The backend streams the simulated clock, pause state, queue length, and order velocity to the frontend via SSE. The frontend can use this data to animate order cards, display live kitchen load, and show current inventory, supplier reliability, and staff capacity which are all ready for integration.

- **Reasoning Log Panel (Thought Trace):** A transparent UI displaying the real-time P-Agent vs. R-Agent debate process and timestamped decision resolutions.

- **"God Mode" Console:**
  Real-time sliders that let you modify:
  - Order generation speed
  - Inventory quantity (globally or per item)
  - Inventory unit cost
  - Oil price

  Note: Currency exchange rate and inflation sliders are reserved for future versions.

- **Human Authorization & Decision Panel:** Administrative safeguard to approve/reject high-impact agent suggestions.

- **Multi-Graph Agent Architecture:** Four dedicated LangGraph sub-graphs (Conversation + Debate, Demand Forecasting, Proactive Crisis, Invoice Ingestion) run independently and coordinate through the shared database.

- **Time Simulation Controls:** Pause / Resume the simulated clock. The engine advances 30 minutes of restaurant time for every 2 seconds of real time (configurable). All business logic — inventory consumption, order generation, expiry risk — follows this

simulated timeline. The simulation starts automatically with the system; no manual "Start" button is required.

7. Features Not Included (Scope Control)

- **Physical POS Hardware Integration:** The system will not integrate with physical receipt printers or legacy local terminal hardware.
- **Live Payment Gateways:** Actual financial transaction processing (e.g., Stripe, PayPal) is excluded; the system assumes transactions are completed for simulation purposes.
- **Automated External Messaging Execution:** While the system will draft supplier WhatsApp messages or emails, it will not autonomously send them to external real-world servers (requires human 'Send' authorization).
- **Multi-Tenant SaaS Infrastructure:** The database schema and application are scoped to a single-restaurant environment to demonstrate the core kernel for the hackathon.

8. Assumptions & Constraints

- **LLM Cost Constraint:** Running complex Dual-Agent debates on every minor customer order is financially unscalable.

  Design Decision: The agentic debate is triggered by:

  - user chat keywords
  - invoice price spikes >20%
  - backend-detected anomalies (stock < 1 day, order surge).

  The system also uses a RAG Knowledge Base to retrieve similar past decisions (e.g., generate_post_mortem_learning), enriching the current prompt with historical lessons – not to bypass LLM calls entirely, but to improve decision quality.

- **Technical Constraints:** The system relies heavily on Supabase Realtime websockets. If significant network latency occurs, the frontend "live sync" of the menu and order queues may experience a marginal 1-2 second synchronization delay.

- **Performance Constraints:** High-quality agentic reasoning requires processing time (typically 3-5 seconds per complex GLM resolution). The UI must rely on robust loading states (e.g., "Agent is thinking...") to manage user expectations during these asynchronous operations.

- **User Input Constraints:** The system assumes that human managers will actively monitor the Human Decision Panel. If critical alerts are ignored, the system is constrained to halt risky execution pathways to prevent operational damage.

9. Risks & Questions Throughout Development

- **Strategic Repetition (Design Similarity):** How do we ensure the AI doesn't monotonously apply a generic "20% discount" to every inventory problem?
    - **Mitigation:** The "Evolution Phase" utilizes `generate_post_mortem_learning`. The AI compares expected yield vs. actual yield post-action, writing the "Lesson Learned" into the RAG vector database. This prevents repeated failures and encourages diverse strategies like dynamic bundling.

- **Financial Hallucinations (Frame Stability):** What happens if the AI hallucinates and prices premium Wagyu beef at RM1.00?
    - **Mitigation:** Strict backend boundary constraints defined in the agent_permissions table: maximum spend without approval = RM500, maximum price change = 15%, maximum discount = 30%. Any action that exceeds these limits automatically generates a pending_approval notification and halts execution until a human approves.

- **Unstructured Input Failure (Drawing Interpretation):** What if the uploaded supplier invoice is entirely illegible?
    - **Mitigation:** If the OCR extraction yields confidence scores below the functional threshold, the Agent gracefully aborts the automated mapping. It falls back to presenting the manager with a manual CRUD entry form pre-filled with whatever partial data was salvageable.

10. Success Metrics

To validate the impact of the AutoYield Kernel, the following metrics will be measured against traditional, non-agentic operational baselines:

- **Autonomous Resolution Rate:** The percentage of detected inventory or pricing anomalies successfully resolved by the AI without requiring human intervention.
- **Time Saved per Incident:** The reduction in time taken to process invoices and execute menu updates (Measured in seconds via AutoYield vs. estimated minutes/hours manually).
- **Simulated Yield Uplift:** The total percentage of profit margin retained during a "God-Mode Crisis Injection" (e.g., a massive supply shock) compared to a static pricing baseline.
- **Back-of-House Waste Reduction:** The simulated decrease in expired inventory volume achieved through proactive, auto-triggered promotional liquidation.